

Project Design Phase

Solution Architecture

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Team ID: SWTID-2026-5429

Project Name: UCab – MERN Stack Cab Booking System

Solution Architecture

Introduction

The **UCab** project follows a **three-tier MERN architecture**, which separates the application into three distinct layers: the Presentation Layer (Frontend), Application Layer (Backend), and Data Layer (Database). This architecture ensures modularity, scalability, maintainability, and secure communication between different components of the application.

The frontend is responsible for interacting with users through a responsive web interface. The backend processes user requests, manages business logic, authenticates users, and communicates with the database. MongoDB stores all application data, including user information and booking details.

This architecture allows the application to handle user authentication, cab booking, fare estimation, booking history, and profile management efficiently.

Objectives of the Solution Architecture

The primary objectives of the UCab solution architecture are:

- To provide a secure and reliable online cab booking platform.
- To ensure efficient communication between the frontend and backend using RESTful APIs.
- To securely store user and booking information in MongoDB.
- To implement authentication and authorization using JSON Web Tokens (JWT).
- To develop a responsive and user-friendly web application using React.js.
- To provide a scalable architecture that supports future enhancements such as live GPS tracking, payment gateways, and driver/admin modules.

Architecture Overview

The UCab application consists of the following layers:

1. Presentation Layer (Frontend)

The Presentation Layer is responsible for displaying information to users and collecting user inputs.

Technologies Used

- React.js
- Vite
- Bootstrap 5
- HTML5
- CSS3
- JavaScript (ES6)
- Axios
- React Router DOM

Responsibilities

- User Registration
- User Login
- Cab Booking
- Fare Estimation
- Booking History
- Profile Management
- Responsive User Interface

2. Application Layer (Backend)

The Application Layer contains the business logic and processes requests received from the frontend.

Technologies Used

- Node.js
- Express.js
- JWT Authentication

- bcryptjs
- Mongoose

Responsibilities

- User Authentication
- Authorization
- Booking Management
- Fare Calculation
- CRUD Operations
- API Development
- Request Validation
- Error Handling

3. Data Layer (Database)

The Data Layer stores and manages all application data.

Technology Used

- MongoDB
- Mongoose ODM

Collections

- Users
- Bookings
- Rides

System Workflow

The UCab system operates through the following workflow:

1. The user opens the UCab application through a web browser.
2. The React frontend displays the Home page and allows the user to register or log in.
3. User credentials are sent to the Express backend using Axios.
4. The backend validates the credentials and authenticates the user using JWT.

5. After successful authentication, the user accesses the dashboard.
6. The user enters pickup and destination locations.
7. The backend calculates the estimated fare.
8. Upon booking confirmation, booking details are stored in MongoDB.
9. The user can view booking history and profile information.
10. The application returns responses to the frontend through REST APIs.

Data Flow

The overall data flow in UCab is as follows:

User → React Frontend → Axios → Express Server → Business Logic → MongoDB Database → Express Server → React Frontend → User

This communication follows a client-server architecture where all data exchange occurs through secure REST APIs.

Security Architecture

UCab implements several security mechanisms to protect user information.

Authentication

- JSON Web Token (JWT)

Password Security

- Passwords are encrypted using bcryptjs before storing them in MongoDB.

Authorization

- Protected routes ensure only authenticated users can access secure resources.

Input Validation

- User input is validated before processing requests.

API Communication

The frontend communicates with the backend using Axios.

The backend exposes RESTful APIs for:

- User Registration

- User Login
- Booking Management
- User Profile
- Booking History

The APIs exchange data in JSON format.

Database Design

MongoDB stores all application information using collections.

Users Collection

Stores:

- Name
- Email
- Password
- Profile Information

Bookings Collection

Stores:

- Pickup Location
- Destination
- Cab Type
- Fare
- Booking Date

Rides Collection

Stores:

- Ride Details
- Booking Status
- User Reference

Technology Stack

Frontend

- React.js
- Vite
- Bootstrap
- HTML5
- CSS3
- JavaScript
- Axios

Backend

- Node.js
- Express.js
- JWT
- bcryptjs

Database

- MongoDB
- Mongoose

Development Tools

- Visual Studio Code
- ThunderClient
- Git
- GitHub

Architecture Diagram

UCab – MERN Stack Cab Booking System

Solution Architecture

